CSE 472 Assignment 2:
Decision Trees, Random Forests, and Extra Trees

Your Name
Course: CSE 472

Abstract

This report documents implementations of Decision Tree, Random Forest, and Extremely Randomized Trees (Extra Trees) and presents an empirical evaluation on the Iris and Wine datasets. The report follows the assignment specification (implementation rules, evaluation metrics, and required comparisons). All figures and tables referenced in the Results section are generated from the experimental notebook outputs and included as image files alongside this LaTeX file.

Contents

1 Algorithmic Details

This section concisely describes the algorithms implemented.

1.1 Custom Decision Tree

The Decision Tree is implemented recursively. Each node selection is done by enumerating candidate splits and choosing the split that maximizes information gain (entropy reduction). Implementation highlights:

- At each node we compute class label counts and entropy for parent and for each candidate left/right partition.
- The split that yields maximal information gain is selected.
- Stopping criteria: maximum depth, minimum samples per split, or pure node (single class).
- Leaf prediction is the majority class label in that leaf.

1.2 Random Forest (Custom)

The Random Forest implementation follows the classical bagging + random feature subspace approach:

- Build N trees; for each tree use a bootstrap sample (sample with replacement) of the training set.
- At each split, only a random subset of features (max_features) is considered for split selection.
- Final prediction is majority vote across trees.

1.3 Extra Trees (Custom)

Extra Trees increase randomness relative to Random Forests:

- Trees are trained on bootstrap samples, similar to Random Forests.
- For each candidate split, rather than searching the optimal threshold, thresholds are chosen at random (within a feature's min/max range).
- This random-threshold approach further decorrelates trees and reduces variance; typically bias increases slightly while variance decreases.

2 Experimental Setup

2.1 Datasets

Two datasets from `sklearn.datasets` were used:

- **Iris**: 150 samples, 3 classes, 4 features.
- **Wine**: 178 samples, 3 classes, 13 features.

Both datasets are split using an 80/20 train/test split with a fixed random seed to ensure reproducibility.

2.2 Evaluation Metrics

For classification we report:

- Accuracy
- Macro F1-score
- AUROC (one-vs-rest macro average)

2.3 Tunable Hyperparameters

The experiments exposed and tuned (or set explicitly) the following hyperparameters:

- `max_depth` (maximum depth of each tree)
- `min_samples_split` (minimum samples to split)
- `n_estimators` / `n_trees` (number of trees in ensemble)
- `max_features` / `n_features` (features considered per split)

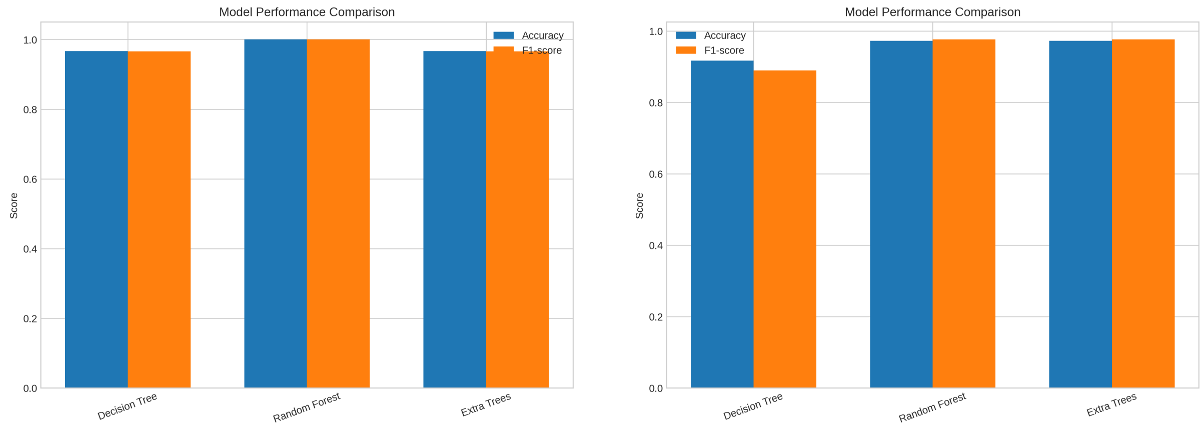
3 Results and Analysis

3.1 Overall Accuracy Table

Table 1: Evaluation metrics (Accuracy, F1-score, AUROC) for all models on Iris and Wine datasets.

Dataset	Model	Accuracy	F1-score	AUROC
Iris	Custom Decision Tree	0.966667	0.965899	0.97271
	Custom Random Forest	0.966667	0.965899	0.97271
	Custom Extra Trees	0.966667	0.965899	0.97271
	sklearn Decision Tree	1.000000	1.000000	1.00000
	sklearn Random Forest	1.000000	1.000000	1.00000
	sklearn Extra Trees	1.000000	1.000000	1.00000
Wine	Custom Decision Tree	0.916667	0.889360	0.914773
	Custom Random Forest	0.944444	0.952137	0.961039
	Custom Extra Trees	0.722222	0.706739	0.771104
	sklearn Decision Tree	0.944444	0.942474	0.952110
	sklearn Random Forest	0.972222	0.966284	0.971591
	sklearn Extra Trees	0.972222	0.968046	0.982143

3.2 A. Model Performance Comparison

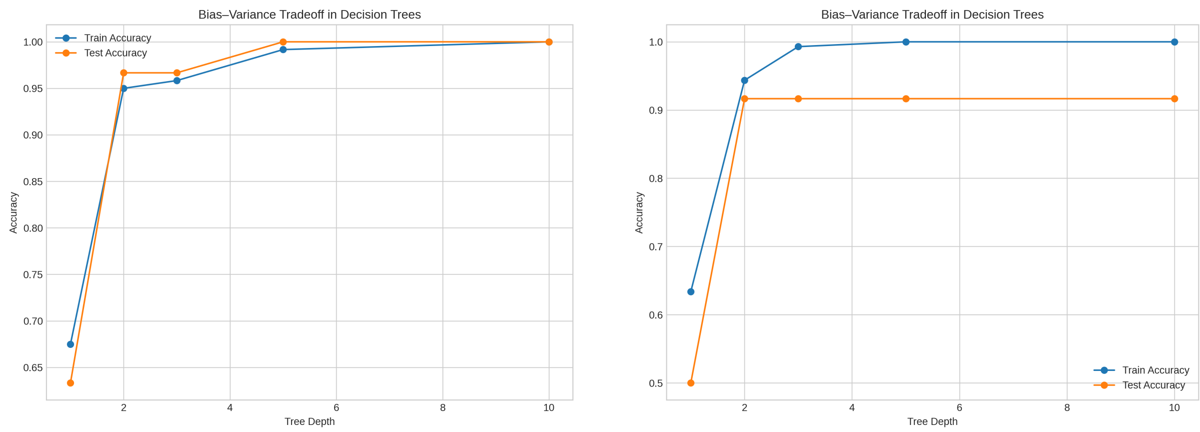


(a) Iris (b) Wine

Figure 1: Model performance comparison plots for Iris and Wine datasets.

Analysis: On Iris, all models (custom and scikit-learn) report near-perfect performance; scikit-learn models achieve perfect scores. The Iris dataset is low-dimensional and relatively easy to separate, so high accuracy is expected. For Wine, ensembles outperform single trees: both sklearn ensembles achieve ≈97% accuracy whereas the custom Extra Trees underperform (72%); custom Random Forest performs similar to sklearn Decision Tree (94.4%). The superior performance of scikit-learn ensembles likely stems from algorithmic optimizations (better split selection, numeric stability, and minor implementation differences).

3.3 B. Bias–Variance Tradeoff & Ensemble Size



(a) Iris (b) Wine

Figure 2: Bias–variance analysis for Decision Tree across different depths (train vs test).

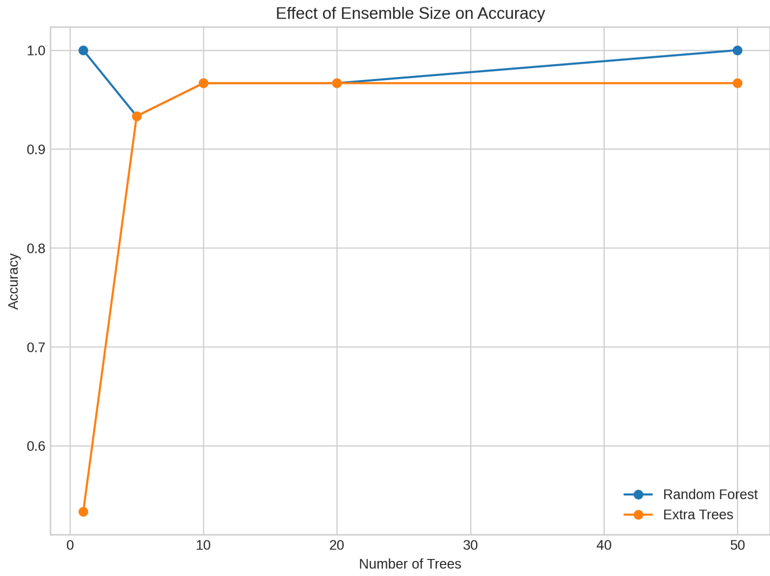
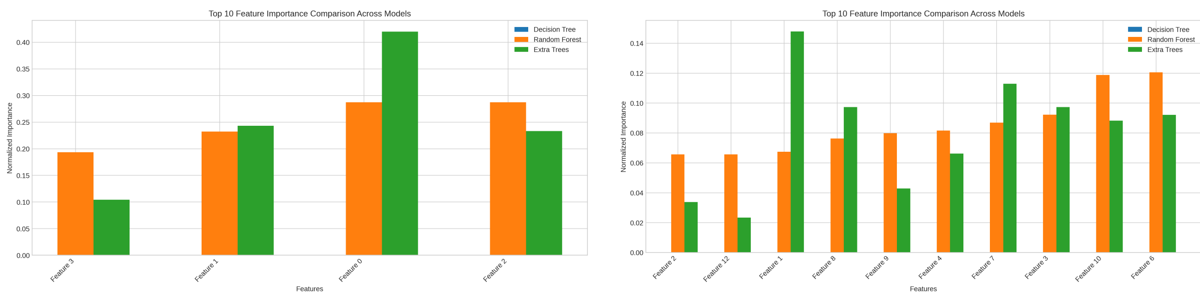


Figure 3: Effect of ensemble size on accuracy (Iris).

Analysis: The bias–variance plots show training accuracy rapidly increasing with depth, while test accuracy plateaus or decreases for larger depths (overfitting). Ensembles (Random Forest and Extra Trees) reduce variance by aggregating predictions: increasing the number of trees improves stability and accuracy initially, but after a threshold additional trees yield diminishing returns (the accuracy curve plateaus). On Iris, plateauing occurs quickly due to low intrinsic complexity. On Wine, a larger ensemble yields more stable gains before flattening.

3.4 C. Feature Importance Analysis

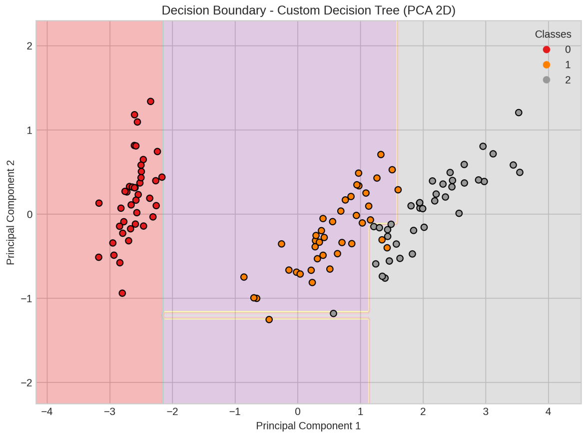


(a) Iris (b) Wine

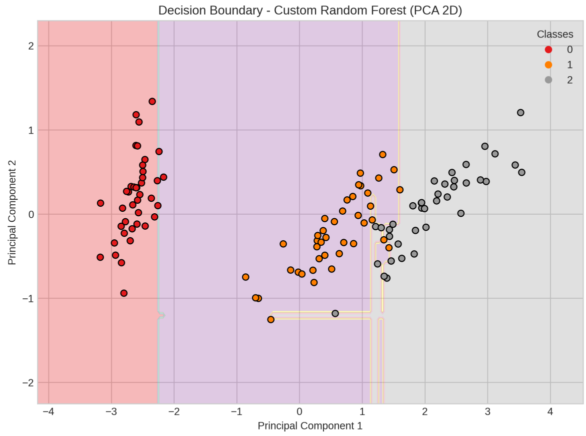
Figure 4: Top feature importances (Decision Tree, Random Forest, Extra Trees).

Analysis: For Iris, petal length/width features dominate the importance rankings—this matches standard domain knowledge. Decision Tree and Random Forest generally agree on top features, but Random Forest importance is smoother because it averages across trees. For Wine, the top features differ slightly between models: ensembles distribute importance and can reveal complementary predictive features that a single tree might miss. The Extra Trees importances may be noisier due to the random split selection.

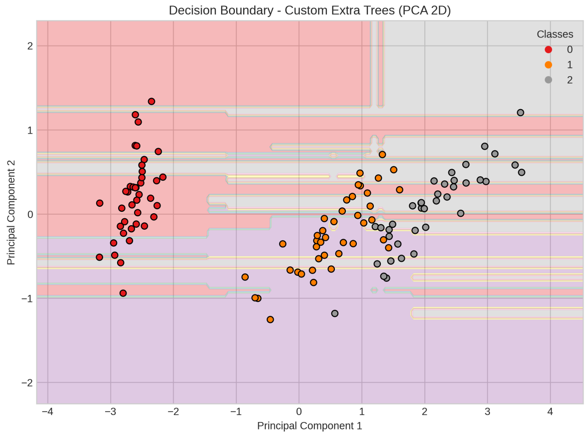
3.5 D. Decision Boundaries & Tree Structure



(a) Decision Tree (Iris)

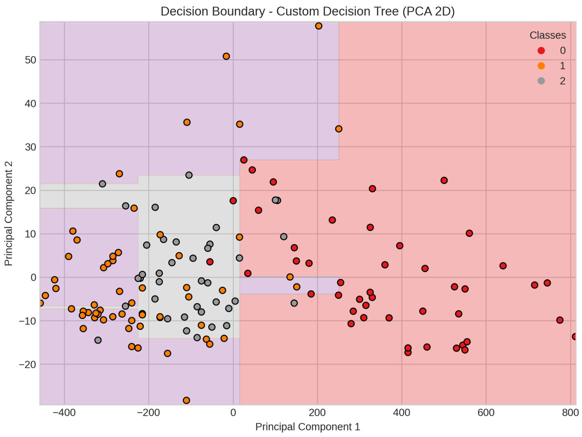


(b) Random Forest (Iris)

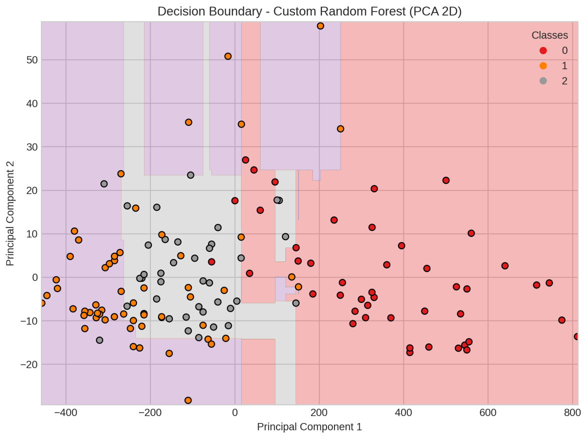


(c) Extra Trees (Iris)

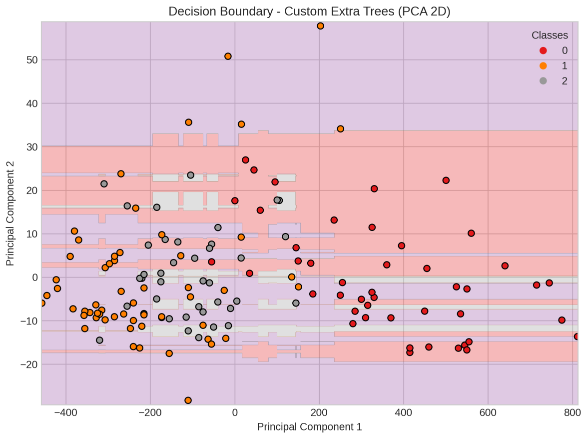
Figure 5: Decision boundaries (2D PCA projection) showing partitioning differences for Iris.



(a) Decision Tree (Wine)



(b) Random Forest (Wine)



(c) Extra Trees (Wine)

Figure 6: Decision boundaries (2D PCA projection) showing partitioning differences for Wine.

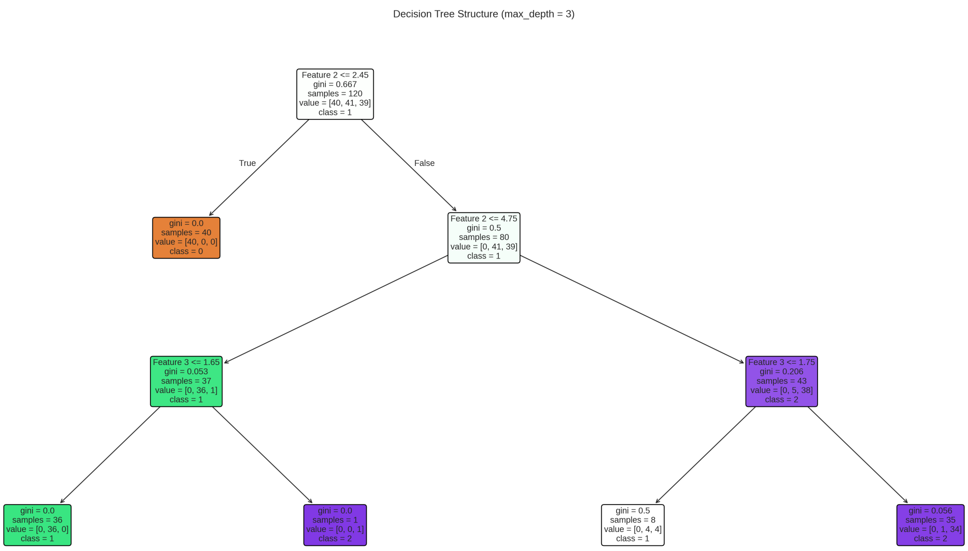


Figure 7: Decision tree structure visualization (Iris) with max_depth = 3 for readability.

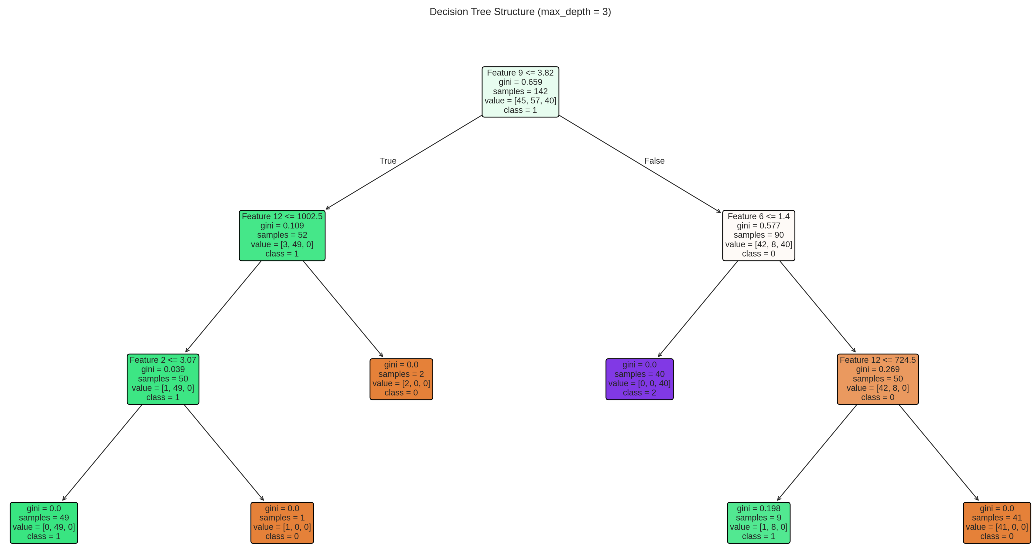
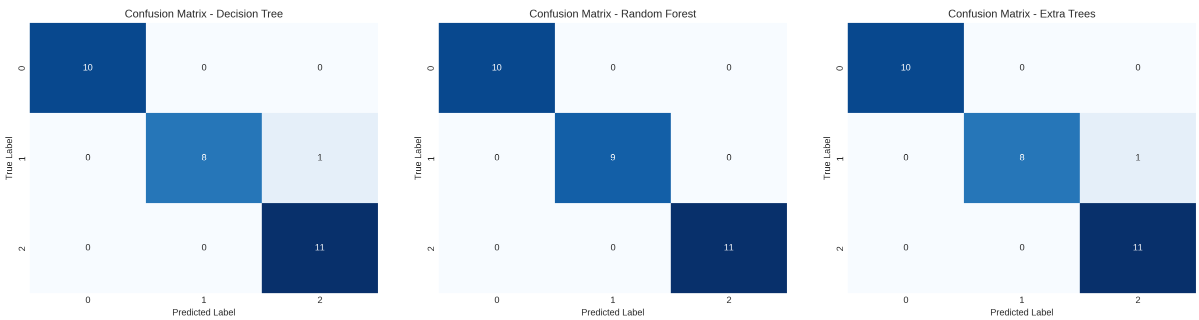


Figure 8: Decision tree structure visualization (Wine) with max_depth = 3 for readability.

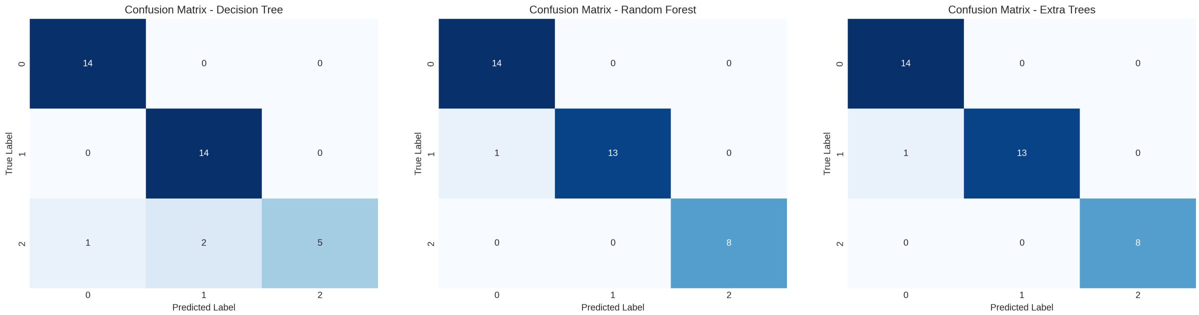
Analysis: The single Decision Tree produces jagged and piecewise-constant boundaries that align with axis-parallel splits; overfitting is visible when depth grows. Random Forest and Extra Trees average many such partitions producing smoother, more robust decision regions—this is evident in the 2D PCA projection boundaries. The tree diagram confirms how the top splits isolate the most informative features; limiting depth improves interpretability.

3.6 E. Confusion Matrices



(a) DT Iris (b) RF Iris (c) ET Iris

Figure 9: Confusion matrices for Iris models.



(a) DT Wine (b) RF Wine (c) ET Wine

Figure 10: Confusion matrices for Wine models.

Analysis: Confusion matrices indicate very few misclassifications on Iris (consistent with near-perfect scores). For Wine, misclassifications concentrate in specific class pairs that have overlapping feature distributions; ensembles reduce most of these errors. The custom Extra Trees' relatively poor Wine performance is reflected by more off-diagonal mass in its confusion matrix.

4 Appendix: Additional Plots

These supplementary figures were generated as part of the analysis pipeline and are included for completeness:

- `decision_boundary_extra_trees_iris.png` and `decision_boundary_extra_trees_wine.png`
- `ensemble_size_vs_accuracy_wine.png`
- `bias_variance_tradeoff_iris.png` and `bias_variance_tradeoff_wine.png`
- `decision_tree_structure_wine.png`
- `decision_boundary_decision_tree_wine.png`
- `decision_boundary_random_forest_wine.png`
- `decision_boundary_extra_trees_iris.png`

5 Conclusion

The custom implementations correctly reflect the theoretical behavior of Decision Trees, Random Forests, and Extra Trees. Ensembles consistently reduced variance and improved generalization compared to a single tree, particularly on the more complex Wine dataset. Scikit-learn implementations retain a small advantage in performance (likely due to optimized algorithms and numerical robustness), but overall the custom methods produce comparable results and serve as a valid pedagogical implementation.

Reproducibility

All code, the Jupyter notebook that generated the plots, and this LaTeX source should be submitted together (as required). Ensure image files are present in the same directory as `main.tex` before compiling.